

SOC ANALYST PORTFOLIO CASE STUDY

Detecting a Web Directory-Traversal Attempt

Wazuh investigation of a controlled path-traversal request in a virtualized home lab

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OUTCOME Generated a controlled Apache path-traversal request, verified HTTP 400 prevented file exposure, and investigated a Wazuh level 6 common-web-attack alert mapped to discovery and public-facing application techniques.

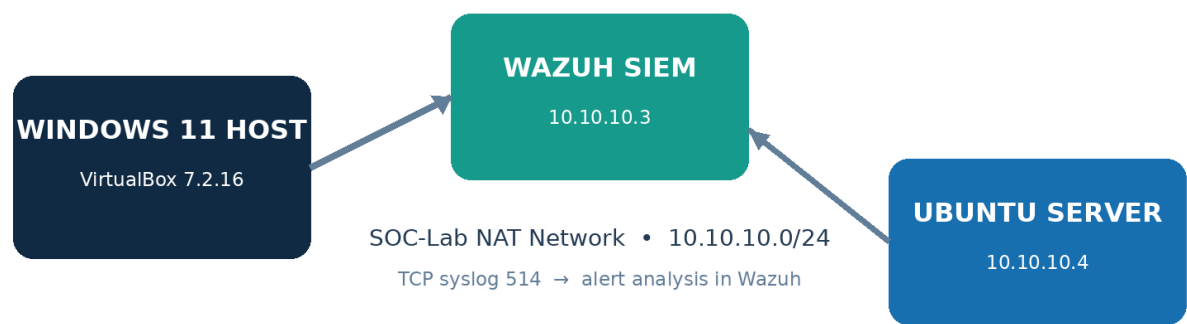


Figure 1. Home-lab architecture and Apache telemetry flow.

Portfolio note: Authorized local simulation only. Apache rejected the request, protected file contents were not exposed, and no production or third-party system was targeted. AI assistance is disclosed within this report.

1. Executive Summary

I tested Wazuh's ability to recognize path-traversal indicators in Apache access logs. After confirming network connectivity and the Apache/rsyslog pipeline, I submitted one request containing repeated ../ sequences to the local server at 127.0.0.1. The request referenced /etc/passwd, a common traversal target, while curl preserved the raw path for accurate detection testing.

Apache returned HTTP 400 and displayed no file contents. Wazuh decoded the access record, extracted the source, method, response, and URL, then raised rule 31104 at level 6 with the description Common web attack. The alert included MITRE context for File and Directory Discovery and Exploit Public-Facing Application.

Project dimension	Implementation
Role performed	Detection tester, log analyst, and Tier 1 incident analyst
Telemetry source	Apache access log from linux-webserver01
Test action	One local GET request containing ../../../../etc/passwd
Preventive result	Apache HTTP 400; no protected file content returned
Detection result	Wazuh rule 31104; level 6; Common web attack
Relevant MITRE context	T1083 File and Directory Discovery; T1190 Exploit Public-Facing Application
Disposition	True positive detection; expected authorized lab activity

2. Objectives

- Validate the existing Apache-to-Wazuh telemetry path after restart.
- Generate a safe, uniquely identifiable directory-traversal indicator.
- Confirm the server rejected the request and protected data remained inaccessible.
- Investigate decoded HTTP fields, rule metadata, and MITRE mappings.
- Document analyst judgment instead of treating every rule mapping as proven behavior.

3. Lab Scope and Controls

Control	Implementation
Authorized target	Local Apache instance on linux-webserver01
Request source	127.0.0.1; no external traffic
Network segment	SOC-Lab NAT network 10.10.10.0/24
Telemetry transport	rsyslog TCP 514 to Wazuh 10.10.10.3
Safety condition	Single request; HTTP 400 required; stop if file content appeared
Evidence marker	SOC-Lab-Path-Traversal

4. Implementation Workflow

01 CONNECTIVITY VALIDATION

Confirmed four successful ICMP replies from 10.10.10.3 with 0% packet loss.

02 SERVICE VALIDATION

Confirmed Apache and rsyslog were active and Apache returned HTTP/1.1 200 OK for the default page.

03 CONTROLLED REQUEST

Used curl --path-as-is so the ../ segments were sent without client-side normalization.

04 SAFETY VERIFICATION

Observed Apache return HTTP 400 Bad Request and confirmed that /etc/passwd contents were not displayed.

05 LOCAL EVIDENCE

Verified the exact URL and SOC-Lab-Path-Traversal marker in /var/log/apache2/access.log.

06 SIEM INVESTIGATION

Filtered Wazuh Discover for the unique marker, expanded the event, and recorded the rule, decoder, fields, severity, MITRE context, and compliance tags.

5. Controlled Request

```
curl --path-as-is -A "SOC-Lab-Path-Traversal" \  
"http://127.0.0.1/../../etc/passwd"
```

SAFETY BOUNDARY The command generated one detection-oriented HTTP request. It did not modify the server, execute code, or obtain protected file contents.

6. Evidence: Request and Server Response

The terminal evidence records the exact request, the HTTP 400 response, and the matching Apache access-log entry. These observations establish both the attempted behavior and the absence of successful file disclosure.



Figure 2. Controlled traversal request, HTTP 400 response, and Apache access record.

Evidence	Observation
Requested path	../../../../etc/passwd
User-agent marker	SOC-Lab-Path-Traversal
HTTP response	400 Bad Request
Protected content	Not returned
Log source	/var/log/apache2/access.log
Source address	127.0.0.1

7. Evidence: Event-Level Investigation

Wazuh decoded the forwarded message as a web access log and preserved the raw traversal sequence. The event identifies the endpoint, local source, HTTP method, response code, and exact URL needed for analyst triage.

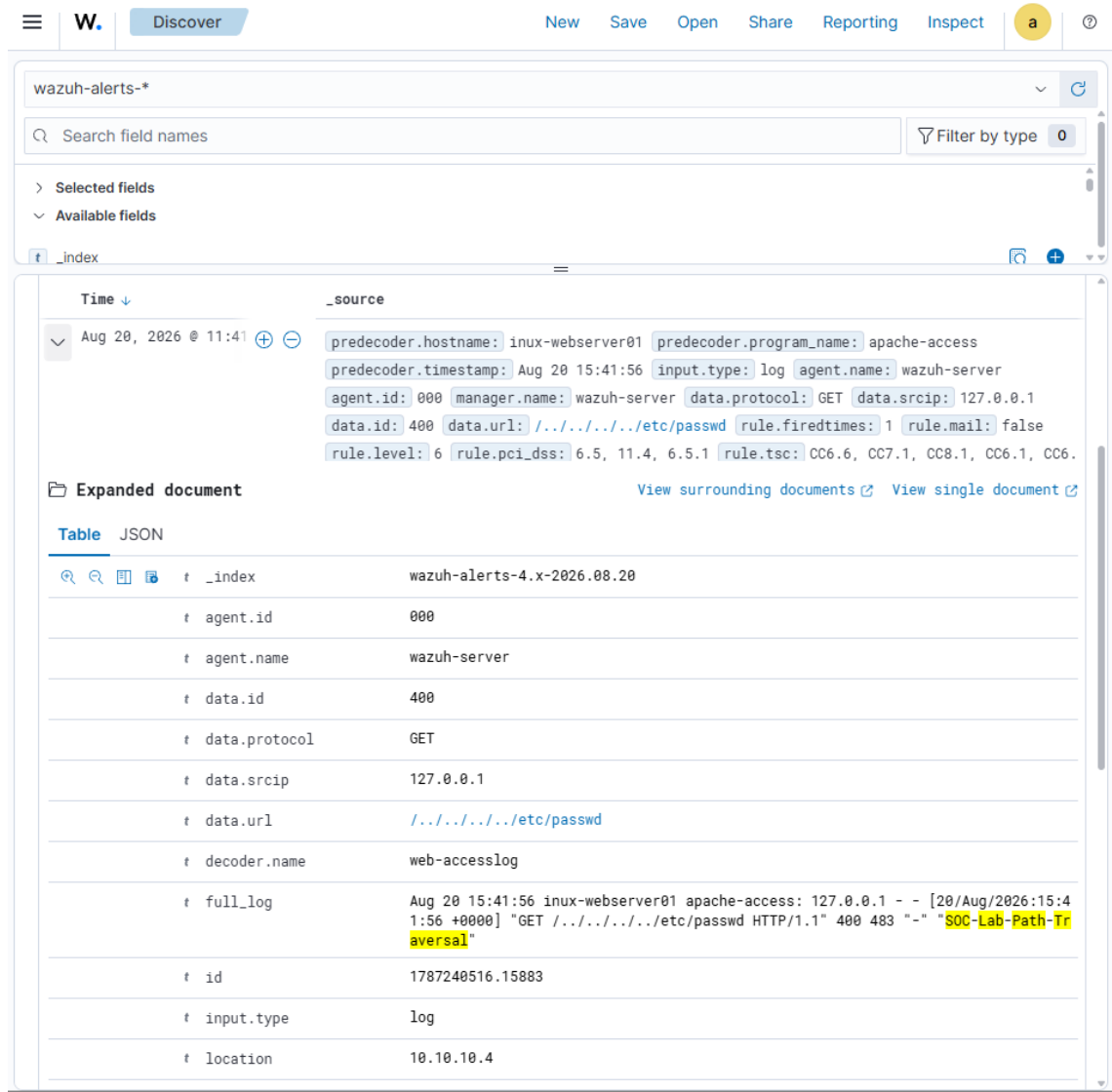


Figure 3. Expanded Wazuh event with decoded traversal and HTTP fields.

Field	Observed value
Endpoint / location	linux-webserver01 / 10.10.10.4
Program / decoder	apache-access / web-accesslog
Source IP	127.0.0.1
HTTP method / result	GET / 400
URL	/../../../../etc/passwd
Rule	31104 — Common web attack
Severity	Level 6
Groups	web, accesslog, attack

8. MITRE ATT&CK and Compliance Context

t	manager.name	wazuh-server
t	predecoder.hostname	linux-webserver01
t	predecoder.program_name	apache-access
t	predecoder.timestamp	Aug 20 15:41:56
t	rule.description	Common web attack.
#	rule.firedtimes	1
t	rule.gdpr	IV_35.7.d
t	rule.groups	web, accesslog, attack
t	rule.id	31104
#	rule.level	6
🔍 🔍 📄 📄 🔍	rule.mail	false
t	rule.mitre.id	T1055, T1083, T1190
t	rule.mitre.tactic	Defense Evasion, Privilege Escalation, Discovery, Initial Access
t	rule.mitre.technique	Process Injection, File and Directory Discovery, Exploit Public-Facing Application
t	rule.nist_800_53	SA.11, SI.4
t	rule.pci_dss	6.5, 11.4, 6.5.1
t	rule.tsc	CC6.6, CC7.1, CC8.1, CC6.1, CC6.8, CC7.2, CC7.3
📅	timestamp	Aug 20, 2026 @ 11:41:56.002

Figure 4. Wazuh rule, MITRE ATT&CK, and compliance metadata.

Mapping	Analyst interpretation
T1083 — File and Directory Discovery	Directly relevant: the request attempted to reference a sensitive filesystem path.
T1190 — Exploit Public-Facing Application	Relevant behavioral context: the request used a web service as the entry point.
T1055 — Process Injection	Rule metadata only; no process injection evidence was observed in this lab.
Compliance context	NIST SA.11/SI.4, PCI DSS 6.5/11.4/6.5.1, and TSC control tags support categorization but do not prove impact.

ANALYST JUDGMENT Rule metadata guides investigation. It should not be reported as proof that every associated technique occurred.

9. Triage and Final Disposition

Question	Assessment
What happened?	One local HTTP request contained repeated parent-directory traversal sequences.

Question	Assessment
What was targeted?	The Apache server and the pathname /etc/passwd.
Was access successful?	No. Apache returned 400 and exposed no file contents.
Why did Wazuh alert?	Rule 31104 matched ../../ as a common web-attack indicator.
What was the scope?	One controlled request from 127.0.0.1.
Classification	True positive detection; expected benign activity in an authorized lab.

FINAL DISPOSITION TRUE POSITIVE — EXPECTED LAB ACTIVITY. Detection worked as designed; no exploitation, unauthorized access, or data exposure occurred.

10. Recommended Response in a Real Environment

- Validate the source IP, requested path, response code, user-agent, and affected application owner.
- Search for repeated traversal variants, encoded paths, successful 2xx responses, and follow-on requests.
- Review WAF, reverse-proxy, application, endpoint, authentication, and file-access telemetry.
- Block or rate-limit confirmed hostile sources and deploy path-normalization/WAF protections.
- Verify web-server document-root restrictions, operating-system permissions, patch level, and least privilege.
- Preserve evidence and escalate if protected content, code execution, persistence, or lateral movement is observed.

11. Troubleshooting and Engineering Judgment

Decision	Reason
Used --path-as-is	Prevented curl from normalizing away the traversal sequence before transmission.
Required a 400/403/404 outcome	Ensured the test stopped without exposing protected content.
Used a unique user-agent	Allowed precise correlation across the access log and Wazuh index.
Separated rule mappings from evidence	Avoided claiming process injection or privilege escalation without supporting telemetry.

12. Assistance and Source Transparency

This project was completed by Ryan Bryant in his personal home lab with assistance from OpenAI ChatGPT (“Sarah”). ChatGPT helped plan the safe test, interpret results, identify validation steps, locate documentation, and edit this portfolio narrative. Ryan personally operated the lab, entered commands, generated the request, verified the server response, investigated Wazuh, captured the screenshots, and confirmed the findings. ChatGPT did not independently access or operate the lab computer.

13. Skills Demonstrated

- Web access-log analysis
- Directory-traversal detection
- Safe adversary simulation
- Wazuh rule and field interpretation
- MITRE ATT&CK contextual mapping
- Evidence-versus-metadata distinction
- Incident triage and disposition
- Response recommendation development
- Ethical scope control

14. Employer-Facing Project Summary

RESUME BULLET Generated and investigated a controlled web directory-traversal event in a Wazuh SIEM lab, validating Apache/rsyslog telemetry, rule 31104 at level 6, HTTP 400 prevention, and MITRE ATT&CK T1083/T1190 context.

Interview talking points

- Why curl path normalization had to be disabled for accurate testing.
- How I proved the request was detected but not successful.
- Why rule metadata is investigative context rather than proof of every technique.
- What additional WAF and endpoint telemetry I would collect in production.

15. References and Validation Sources

Official documentation and observed system evidence were used to validate the test design and analysis.

[OpenAI key concepts](#)

[Wazuh log data analysis](#)

[Wazuh web-access rules](#)

[Apache access-log documentation](#)

[MITRE ATT&CK T1083](#)

[MITRE ATT&CK T1190](#)